

Scope of Work (Development Document)

GeoTracker

AI Visibility Audit Platform — Technical Build Plan

Project	GeoTracker
Parent Company	lozera
Derived From	GeoTracker BRD v1.0
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This Scope of Work translates BRD v1.0 into an actionable engineering plan. It defines deliverables, milestones, acceptance criteria, technical architecture, and exit conditions. Where the BRD describes *what* and *why*, this document defines *how* and *when*. The BRD remains the contract; the SOW is the build plan.

1. Purpose & Scope

Build and deploy **GeoTracker**, an AI visibility audit web application. Users paste a business domain and instantly see whether it is recommended by ChatGPT, Perplexity, Claude, Gemini, and Grok for buying-intent prompts across their service catchment area. The app functions as the primary lead generation channel for Upserv.ai and as a freemium SaaS product in its own right.

1.1 In Scope (Phase 1)

- Web application: hero, audit engine, reveal dashboard, vendor comparison, CTA flow.
- Async parallel querying of 5 LLM APIs (ChatGPT, Perplexity, Claude, Gemini, Grok).
- Tiered Green / Yellow / Red scoring with contact-info downgrade rule.
- Geospatial catchment expansion using a business-type radius lookup (20 industries seeded).
- OAuth (Google + LinkedIn), free + premium tiers, monthly Pulse Report email job.
- Admin dashboard: lead tracking, user management, API health, conversion metrics.
- Real-time progress bar via SSE during audit execution.
- Dental vertical prompt set at full depth; 12+ generic verticals at fallback depth.

1.2 Out of Scope (Deferred to Phase 2+)

- Competitive Conquest mode (user adds competitors for head-to-head audits).
- Predictive Visibility analytics (training-trend signals).
- White-label / agency reseller portal.
- Public API for power users and integrators.
- LLM coverage beyond the Phase-1 five.
- Native mobile applications.
- Verticalized prompt depth for the 12+ generic industries (content task, not engineering).

1.3 Explicitly Excluded

- Upserv.ai delivery (the ongoing red-X remediation service) is a separate workstream owned by the Upserv team.
- Kriss.ai chatbot integration into GeoTracker — handled in a future cross-product epic.
- Sales rep CRM tooling — reps use existing IoZera workflow.

2. Assumptions, Constraints, Dependencies

2.1 Assumptions

- API access available for all 5 LLMs at developer tier with reasonable rate limits.

- Google Maps Platform billing enabled for Geocoding + Places API.
- OAuth credentials provisioned for Google and LinkedIn ahead of integration milestone.
- Dental vertical is the launch market; other verticals follow.
- Sales rep distribution channel exists (chit-chat outreach per BRD §10).

2.2 Constraints

- Total audit latency must remain under 15 seconds (BRD §7.2).
- Phase 1 must ship to a hosted environment, not localhost-only.
- Architecture must support 1,000+ concurrent reps without degradation.
- All LLM API keys, OAuth secrets, and email credentials must live in a secrets manager — never client-side.

2.3 External Dependencies

Dependency	Owner	Required By
OpenAI API key	Tim / lozera	LLM integration milestone
Perplexity API key	Tim / lozera	LLM integration milestone
Anthropic API key	Tim / lozera	LLM integration milestone
Google Gemini API key	Tim / lozera	LLM integration milestone
xAI (Grok) API key	Tim / lozera	LLM integration milestone
Google Maps API key	Tim / lozera	Geospatial milestone
Google + LinkedIn OAuth credentials	Tim / lozera	Auth milestone
Transactional email account	Allan	Pulse Report milestone
Railway / Fly.io account for API host	Allan	Deploy milestone
Vercel account for frontend	Allan (existing)	Deploy milestone
Token budget approval (real-LLM flip)	Tim	Production launch

3. Technical Architecture

3.1 High-Level Topology

GeoTracker is a two-tier system: a static Astro/React frontend hosted on Vercel, and a long-running Fastify API hosted on a container platform (Railway recommended). The API orchestrates LLM calls asynchronously via BullMQ

on Redis, persists audits to Postgres, and streams real-time progress to the client via Server-Sent Events.

3.2 Stack — Confirmed

Layer	Technology	Notes
Frontend	Astro 5 + React + Tailwind + shadcn/ui	House lozera stack (per Diamond Dental, Upserv).
Backend API	Fastify (Node.js)	Async-first, low overhead, fits long-running process.
Job queue	BullMQ + Redis	Handles 5×N LLM fan-out per audit.
Database	Postgres	Users, audits, lookup tables, Pulse subscriptions.
Cache	Redis	Recent audit deduplication, rate-limit windows.
Real-time	Server-Sent Events (SSE)	Per-LLM progress bar updates.
Auth	OAuth 2.0 (Google + LinkedIn)	Session via signed JWT cookies.
Hosting (web)	Vercel	Static + edge functions.
Hosting (API)	Railway	Managed Postgres + Redis in same project.
Geospatial	Google Maps Geocoding + Places	Mapbox as fallback.
Email	Postmark or Resend	Transactional Pulse Reports.
Analytics	PostHog	Server-side capture on key conversion events.

3.3 Service Modules (Internal)

Note on terminology: BRD §12 refers to capabilities as 'skills.' For Phase 1, these are implemented as **modular service modules** inside the Fastify runtime — not Claude Skill packaging. This is documented as DRIFT-001 in the project status. Rationale: the runtime is Fastify with strict latency requirements; an agent-dispatch layer adds indirection without functional benefit at this layer.

Service Module	Responsibility
audit-orchestrator	Receives audit request, fans out LLM jobs in parallel, emits SSE progress events.
llm-adapter (×5)	Per-provider wrapper: ChatGPT, Perplexity, Claude, Gemini, Grok. Uniform interface.
geospatial-catchment	Given business address + industry, returns list of cities within radius.
prompt-set-resolver	Given industry, returns prompt template with {city} tokens.
parsing-validation	Detects mentions, ranks, caveats; cross-references domain to avoid false positives.
scoring-engine	Applies Green / Yellow / Red logic, including contact-info downgrade rule.

Service Module	Responsibility
pulse-runner	Scheduled job: re-runs audits monthly, formats and sends email reports.
auth-service	OAuth flows, session management, dev-login (gated to non-prod).
admin-api	Lead tracking, user management, API monitor, conversion metrics endpoints.

3.4 Data Model (Initial Schema)

Table	Key Columns	Purpose
users	id, email, oauth_provider, oauth_id, tier, created_at	Identity + subscription tier.
businesses	id, user_id, domain, name, industry, address, lat, lng	Audited businesses (one user may audit many).
audits	id, business_id, score, status, started_at, finished_at	One row per audit run.
audit_results	id, audit_id, llm, prompt, city, mention_state, raw_response	Per-prompt cell in the grid.
industry_radius	industry, default_radius_miles, prompt_set_id	BRD §5.4 lookup table.
prompt_sets	id, industry, prompts[]	Per-industry buying-intent prompts.
pulse_subscriptions	user_id, business_id, active, last_sent_at	Monthly Pulse Report enrollment.
events	id, user_id, event_name, payload, ts	PostHog mirror for internal analytics.

4. Milestones & Deliverables

Phase 1 is broken into 10 milestones plus 2 hardening passes. Each milestone has explicit acceptance criteria and produces a deliverable that can be demoed independently.

#	Milestone	Key Deliverable
M1	Repo scaffold + CI	Monorepo (apps/web, apps/api), lint + test pipeline, deploy preview.
M2	Data model + migrations	Postgres schema, seed data for 20-industry radius table.
M3	Prompt-set resolver + dental vertical depth	Dental prompt set (8+ prompts) + 12 generic fallback sets.
M4	Geospatial catchment service	Address → list-of-cities-in-radius. Cached.
M5	LLM adapter framework (mock)	Uniform adapter interface, mock fixtures for all 5 LLMs, deterministic output.
M6	Audit orchestrator + BullMQ	Domain in → parallel fan-out → results in DB. SSE progress events emitted.

#	Milestone	Key Deliverable
M7	Parsing + scoring engine	Tiered Green/Yellow/Red logic with contact-info downgrade and generic-name guard
M8	Frontend — hero, progress, reveal	Domain input → live progress bar → visibility score gauge + grid.
M9	Vendor comparison + CTA	Comparison table, consultation CTA, post-audit conversion flow.
M10	Auth + free-tier flow + Pulse Report	Google + LinkedIn OAuth, account dashboard, monthly email job.
H1	Hardening pass — bugfixes from gate runs	BullMQ shared connection, SSE unsubscribe, hydration race, IPv6 binding, etc.
H2	Hardening pass — deploy readiness	Boot guards on dev-login, env validation, observability.
M11	Real-LLM flip (gated on Tim)	Mock adapters replaced with live API calls. Token-budget guardrails active.
M12	Admin dashboard	Lead tracking, user management, API monitor, conversion metrics.

5. Acceptance Criteria

Phase 1 is considered **ship-ready** only when all of the following are true:

#	Criterion	How Verified
AC-1	End-to-end audit completes in ≤15 seconds (P95) against real LLMs.	Static load test, 100 audits.
AC-2	Scoring logic correctly handles all 4 cases: top-3, mentioned-but-not-top3, non-mention, Q/A against fixtures.	Unit tests, manual QA against fixtures.
AC-3	Generic-name false-positive guard works (e.g., 'City Dental' in Houston).	Targeted City Dental in Boston.
AC-4	Geospatial catchment returns sane city lists across 20 industries.	Snapshot tests for each industry.
AC-5	Real-time progress bar updates per-LLM as adapters resolve.	Manual QA in 3 browsers.
AC-6	Free user can complete: signup → audit → Pulse Report opt-in → Manual email received.	Manual end-to-end.
AC-7	No dev-login endpoint reachable in production (hard boot guard).	Deploy smoke test.
AC-8	All LLM API failures degrade gracefully to 'unavailable' rather than crashing the audit.	Crash test the audit, provider mid-audit.
AC-9	Admin dashboard shows audits, lead scores, conversion funnel.	Manual QA + screenshot in status update.
AC-10	PostHog events fire on: audit_started, audit_completed, vendor_consultation_verified.	PostHog dashboard verified.

6. Skills, Tools, and Bots

Per Tim's framework (brainstorm session, May 18 2026), a web application is composed of **skills**, **tools**, and **bots**. This section maps BRD §12 to concrete implementations.

6.1 Skills (Service Modules)

Implemented as Fastify service modules. See §3.3 for the full list. Pending Tim's decision on DRIFT-001, these may be re-packaged as Claude Skills in a Phase 2 epic if integration with an agent runtime (e.g., Kriss.ai) requires it.

6.2 Tools / External Integrations

- OpenAI API (ChatGPT)
- Perplexity API
- Anthropic API (Claude)
- Google Gemini API
- xAI API (Grok)
- Google Maps Geocoding + Places
- Google OAuth 2.0
- LinkedIn OAuth 2.0
- Postmark or Resend (transactional email)
- Redis (BullMQ + cache)
- Postgres
- PostHog (analytics)

6.3 Bots (Scheduled / Reactive Jobs)

- **Audit Scheduler Bot** — monthly cron job, re-runs audits for Pulse Report subscribers.
- **API Health Bot** — continuous probe of each LLM endpoint; alerts on outages.
- **Lead Routing Bot** — flags low-scoring audits as high-priority Upserv leads in the admin dashboard.

7. Testing Strategy

Layer	Approach	Gate
Unit	Vitest. Coverage focus: parser, scorer, catchment, prompt-resolver.	75%+ on core logic modules.
Integration	Spin up Postgres + Redis in CI. Test orchestrator with mock LLM adapter.	All ACs through AC-4 pass.
E2E	Playwright. Full hero → reveal → CTA flow.	AC-5 + AC-6 pass.
Load	k6 or Artillery. 100 concurrent audits against mock adapter.	P95 latency ≤15s sustained.

Layer	Approach	Gate
Chaos	Kill one LLM adapter mid-audit; verify graceful degradation.	AC-8 passes.
Manual QA	Three-browser pass (Chrome, Safari, Firefox).	Sign-off from Allan.

8. Deployment Plan

8.1 Environments

Environment	Purpose	URL Pattern
Local	Development	localhost:3000 / :4000
Preview (Vercel)	Per-PR previews. Mock LLM adapters. Gated (basic auth or allowlist) until real LLMs ship.	preview.geotracker.iozero.ai
Staging	Pre-prod, real LLMs at low rate limit, internal use only.	staging.geotracker.iozero.ai
Production	Live. Real LLMs at full rate limit. Customer-facing.	geotracker.iozero.ai

8.2 Deploy Sequence

- Provision Railway project: managed Postgres + Redis + Fastify service.
- Set env vars: DATABASE_URL, REDIS_URL, all 5 LLM API keys, OAuth credentials, email API key, PostHog key.
- Deploy API to Railway; verify health endpoint and BullMQ workers running.
- Point Vercel PUBLIC_API_ORIGIN at the Railway URL.
- Run E2E smoke test against staging.
- Gate Vercel preview behind basic auth until Tim approves real-LLM flip.
- On real-LLM flip approval: cut M11; run AC-1 (live latency check) and AC-8 (chaos test).

9. Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
Token budget at scale exceeds Tim's tolerance (\$430K/yr high).	Medium	High	Cap city iterations per audit. Negotiate partner tier with at least 1M requests.
LLM provider TOS issues around programmatically generated queries.	Medium	High	Legal review before M11. Identify any provider that requires human review.
Mock data on public preview poisons sales funnel.	High (if left unmitigated)	High	Gate preview behind basic auth until real LLMs are live. No public access.
Dev-login endpoint accidentally enabled in prod.	Low	Critical	Hard boot guard: throws if NODE_ENV=production and API_KEY is not set.

Risk	Likelihood	Impact	Mitigation
LLM provider rate limiting under 1000-req/day	High	Medium	BullMQ throttling per provider. Distributed request pools. F
Generic prompt fallback hurts conversion in 12 non-dental verticals	Medium	Medium	Ship Phase 1 with dental at depth; verticalize 12 others as
DRIFT-001 escalates to a Phase 1 rework. Low	Low	Medium	Surface to Tim early. Default position: keep services, upda

10. Exit Criteria & Handoff

Phase 1 is complete and the SOW is closed when **all of the following are true**:

- All 10 acceptance criteria (\$5) pass.
- Production deploy on geotracker.iozero.ai is stable for 7 consecutive days.
- Tim has approved the real-LLM flip and signed off on token-budget caps.
- Sales rep playbook is updated to use the live URL in chit-chat outreach.
- PROJECT_STATUS.md final entry written and committed.
- All open DRIFT items either closed or formally moved to Phase 2.

Handoff Artifacts

- Production codebase + tagged v1.0 release.
- Runbook (env vars, deploy commands, incident playbook).
- API documentation (admin endpoints, internal service interfaces).
- Post-launch monitoring dashboard (PostHog + Railway metrics + API health bot).
- Updated BRD with any wording corrections from DRIFT resolutions.

11. Appendix — Claude Code Master Prompt

The canonical kickoff prompt for Claude Code, reproduced from BRD §13 with SOW-level specifics added:

```
Build the GeoTracker web application per SOW v1.0.

Monorepo: apps/web (Astro 5 + React + Tailwind + shadcn/ui) and apps/api (Fastify).
DB: Postgres. Queue: BullMQ on Redis. Real-time: SSE.
Hosting: Vercel (web), Railway (api). Auth: OAuth via Google + LinkedIn.

Implement service modules per SOW §3.3: audit-orchestrator, 5× llm-adapter,
geospatial-catchment, prompt-set-resolver, parsing-validation, scoring-engine,
pulse-runner, auth-service, admin-api.
```

LLM adapters: mock fixtures by default. Real-LLM flip is a separate milestone gated on Tim's approval and token-budget sign-off.

Scoring: Green = top-3 mention. Yellow = mentioned outside top-3 OR no actionable contact info. Red = not mentioned. Apply generic-name guard (cross-ref domain + location).

Geospatial: industry-based radius lookup (20 industries seeded). Iterate {city} token across catchment.

Deliver milestone-by-milestone per SOW §4. After each milestone, update PROJECT_STATUS.md per its own maintenance rules. Surface drift in Section 6 immediately.

12. Sign-Off

Role	Name	Signature / Date
Author	Allan	_____
Approver (CEO)	Timothy Ngo	_____
Engineering Lead	(Claude Code session owner)	_____